MOHAMMED MUDASSIR UDDIN

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SUMMARY

Machine learning engineer and researcher. First author of an ICLR 2027 submission on long-context retrieval in hybrid linear-attention language models. Work spans mechanistic interpretability (scalable circuit extraction, released with a public audit of what the code reproduces), efficient LLM inference (task-aware quantization routing for agents), edge ML (few-shot pruning to 142 ms per image on a Raspberry Pi 4) and a CVPR 2027 submission in preparation on scene-level conformal guarantees for feed-forward 3D reconstruction. Research Intern on WISER, an Indo-German project mapping pest-infested paddy from RGB, thermal and LiDAR imagery, and visiting research collaborator at KFUPM on the faithfulness of clinical AI explanations. Open-source contributor (pgmpy, pyGAM, TransformerLens, nnsight, skpro) and builder of LaTeX Intelligence (VS Code Marketplace).

EDUCATION

Muffakham Jah College of Engineering and Technology, Osmania University Hyderabad, 2023 – 2026
B.E. Computer Science and Engineering | CGPA 7.99/10, First Class with Distinction

State Board of Technical Education and Training (SBTET), Telangana Hyderabad, 2020 – 2023
Diploma in Computer Engineering | CGPA 8.05/10, First Class with Distinction

RESEARCH EXPERIENCE

Research Intern | WISER, Indo-German project, MJCET, with Dr. Tasneem Bano Rehman August 2026 – Present
– WISER is an Indo-German (IGSTC) government-funded project on an AI scout drone for pest detection and geotagging in precision agriculture; I contribute to data acquisition and work with its multimodal data (RGB, thermal, LiDAR).
– My focus is data and sensor fusion: combining the modalities so the project can geotag the field areas most affected by pests.
– Earlier with Dr. Tasneem (2025 – 2026): first author of [3], disease-aware pruning with meta-learning for few-shot plant disease detection on edge hardware.

Visiting Research Collaborator | King Fahd University of Petroleum and Minerals (KFUPM) April 2026 – Remote
– Verifying explanation faithfulness in sepsis clinical decision support with Prof. Hamdi Al-Jamimi: perturb the evidence an explanation names and test whether the recommendation shifts as the explanation predicts.
– Sepsis-3 labels; MIMIC-IV and PhysioNet 2019 first, then HiRID (Swiss ICU) to test whether faithfulness transfers across hospital systems.

Independent project, computer vision | CVPR 2027 submission in preparation 2026 – Present
– Scene-level conformal guarantees for feed-forward 3D reconstruction (VGGT, MapAnything): the scene, not the pixel, is the exchangeable unit, so a chosen fraction of each new scene's pixels stays inside the depth interval, with camera-pose sets and label-free difficulty strata, and no retraining of the model.
– Method designed; evaluation planned on DTU, ETH3D, 7-Scenes and TUM RGB-D against native confidence, power-law recalibration and pixel-level split conformal.

Graduate Research Collaborator | MJCET, with Dr. Fahmina Taranum June 2024 – Present
– Led the ICLR 2027 study [1]: 49 models from 15 hybrid-architecture designs, one mechanism changed at a time, retrieval tested at every depth up to 64× the training length.
– Co-developed two Google TPU Research Awards proposals: TPU-native evidence memory and KV-cache control for long-horizon agents & perturbation-grounded genome modelling in JAX.

PUBLICATIONS

[1] **Layer Order and Key Binding Decide When Hybrid Models Learn to Retrieve** **M. M. Uddin**, F. Taranum, T. B. Rehman, S. Siddiqui, S. Rizwan, S. Abdus Salam, Q. Nayeem, S. U. Begum, S. Kauser, M. Shareef, A. R. Ahmed *(Under review, ICLR 2027)* [**Project Page**]
Global-first block with bound keys: all **11/11** runs learned retrieval by step **300** and kept **98.0%** recall at **64**× the training length; ablating that one layer drops recall from 99.0% to 0.2%.

[2] **Hierarchical Sparse Circuit Extraction from Billion-Parameter Language Models through Scalable Attribution Graph Decomposition** **M. M. Uddin**, S. Alam, M. K. Pasha *(arXiv:2601.12879; Accepted, Discover Artificial Intelligence)* [**Project Page**]
Coarse-to-fine circuit search, $O(n^2 \log n)$ worst case, **0.09 s** over 413 features; released with a five-seed verification suite and a public claim audit of what the code reproduces.

[3] **Meta-Learning Guided Pruning for Few-Shot Plant Pathology on Edge Devices** **M. M. Uddin**, S. Alam, M. K. Pasha, T. B. Rehman, F. Taranum, A. Begum *(arXiv:2601.02353; under review, JMLR)* [**Project Page**]
ResNet-18 from 11.2M to 2.5M parameters (**78%** smaller), **142 ms** per image (**7 FPS**) on a Raspberry Pi 4, **+3.8** points over Meta-Prune at equal size.

[4] **Small Object Detection in Drone Aerial Imagery with LAF-YOLOv10** Q. Nayeem, F. Taranum, **M. M. Uddin** *(arXiv:2609.14560)* [**Project Page**]
Composability stress test: four validated modules combined reach **24.0%** mAP@0.5 on VisDrone against 31.8% for YOLOv10n; the deficit traces to one interaction.

[5] **AgentCompress: Task-Aware Compression for Affordable Large Language Model Agents** Z. A. K. Taha, **M. M. Uddin**, S. Alam *(arXiv:2601.05191)* [**Project Page**]
Per-step FP16/INT8/INT4/INT2 routing for LLM agents; the paper reports **68.3%** less compute at **96.2%** task success on 290 research workflows.

PROFESSIONAL EXPERIENCE

Founder and Head of Research | Auragenzy Hyderabad, Jan 2025 – Feb 2026
– Led the technical and research work for an AI SaaS for no-code workflow automation: the agent orchestration and evaluation stack, and the internal research that decided what to build.
– Took the product from idea to paying customers in under 90 days and reached 200+ active users with no ad spend, shipping a new iteration every two weeks.
– Cut manual task time by up to 70% for early users; open-sourced parts of the model tooling.

Machine Learning Engineer (Intern) | RentEra Remote, Dec 2022 – May 2023
– Refined production ML models on real-estate rental data through feature engineering and algorithm tuning, reducing live inference error on user-facing systems.
– Built models in Python and R alongside senior engineers on user-facing prediction systems.

Junior ML Engineer (Apprentice) | Techieyan Technologies Hyderabad, May 2022 – Nov 2022
– Ran structured exploratory analysis and statistically validated experiments that surfaced the feature signals behind product decisions.
– Improved predictive models through data preprocessing, model development and algorithm implementation; first production ML role, started during the diploma.

PROJECTS AND OPEN SOURCE

LaTeX Intelligence | TypeScript, VS Code API 2026
– LaTeX-to-Word converter with native OMML equations and live cross-reference fields, plus plain-language diagnosis of failed builds; published on the VS Code Marketplace; 397 tests, 153 KB, no runtime dependencies.

Federated Data Node | Python, SQLite, Avro 2026
– Offline-first collection node for Telugu, Tamil, Hindi and Bengali text in which raw text never leaves the device, enforced by four independent egress guards; runs on a 4 GB Raspberry Pi 4 over 2G; 188 tests.

AlphaEarth | Python, Google Earth Engine, gradient-boosted trees 2026
– Parcel-level flood and wildfire damage from two Sentinel-2 passes, scored against independent damage records; 24,608 parcels, leave-one-event-out validation, flood macro-F1 0.906.

Guardian-AI | Python, Streamlit, MITRE ATLAS 2025 – 2026
– Prompt-injection filter for LLMs that read security logs: 34 rules re-run on decoded, homograph-folded and re-joined inputs; caught 12/12 obfuscated attacks on a regression set at 1.1 ms median latency; 113 tests.

Vak | Sarvam AI speech, translation and LLM APIs 2026 – Present
– Open-source AI literacy and interpretability platform built natively in Telugu and Hindi; accepted to the Sarvam AI Startup Program with 25,000 API credits.

Open-source contributions | pgmpy, pyGAM, TransformerLens, nnsight, Flower, datasets, skpro, movement, BrainGlobe 2026
– 11 pull requests to pgmpy (d-separation, inference and model-utility fixes) and 4 to pyGAM (p-values via Wood's rank-r eigendecomposition, link-function overflow, scikit-learn estimator contract).
– Twenty-nine pull requests and ten bug reports since October 2026, each fix with tests: ten merged (TransformerLens, nnsight, HEPData), the rest open at Flower, datasets, skpro, movement, BrainGlobe, zfit, cabinetry, hepstats and Invenio (e.g. batch dimensions in cached hooks, `Hurdle` variance, `create_extended` dropping a PDF's range).

HONORS AND ACHIEVEMENTS

- **Polaris Fellowship (2026):** one of ten fellows selected nationally.
- **Erdős Institute Research Cohort (2026)** and **Sarvam AI Startup Program (2026)**, the latter for Vak.
- **Research proposals under review (2026), PI Dr. Fahmina Taranum:** Indian Council of Social Science Research (ICSSR) minor research project (INR 15 lakh), GARA as Co-PI (circuit-level cost of locally-quantum decoders).
- **Agentathon 2025:** participant in the Guinness World Record event for the largest agentic AI hackathon (2,300+ developers).

CERTIFICATIONS

Neo4j Certified Professional (2026) | Claude Certified Architect, Foundations, Anthropic (2026, 944/1000)

TECHNICAL SKILLS

Research Areas Mechanistic interpretability, long-context architectures, LLM efficiency (quantization, pruning), few-shot learning, multimodal remote sensing (RGB, Thermal, LiDAR), explanation faithfulness, causal inference, uncertainty calibration (conformal prediction), 3D vision
Languages Python, C++, TypeScript, Java, SQL
ML and Systems PyTorch, JAX, TensorFlow, TFLite, Hugging Face Transformers, vLLM, bitsandbytes, GPTQ, LangGraph, Neo4j, Docker, GCP, AWS, Git, CUDA, GPU Programming